Summary of Previous Tested A* Update Strategies and Cost Functions

1 A* Search Cost Components

The A* algorithm evaluates nodes n in the search space using a total estimated cost function $f(n)$, which is typically a combination of the cost-to-come $g(n)$ and the heuristic cost-to-go $h(n)$.

1.1 Heuristic Cost (Cost-to-Go), $h(n)$

The heuristic $h(n)$ represents the estimated cost from the current state (node n) to the goal state. In this context, $h(n)$ is defined as the model's loss on the training data (X, Y):

$$h(n) = \text{Loss}(\text{MLP}_n(X), Y)$$

where MLP_n is the Quantized MLP model associated with node n, and Loss is the chosen loss function.

1.2 Cost-to-Come, $g(n)$

The cost-to-come $g(n)$ is the accumulated actual cost from the initial state n_0 to the current state n. For a transition from a parent node n_{parent} to a neighbor node n, the cost is updated by a step cost Δg:

$$g(n) = g(n_{parent}) + \Delta g(n_{parent} \to n)$$

The initial cost is $g(n_0) = g_{\text{ini}}$ and the step cost Δg is dependent on the chosen update strategy.

1.3 Total Estimated Cost, $f(n)$

The total estimated cost $f(n)$ is calculated based on $g(n)$ and $h(n)$, and may incorporate a scaling factor to dynamically adjust the influence of g versus h.

1.3.1 Standard A* (Unscaled f)

If update_strategy = 0, or if scale_f = False (for strategies 1 and 2), the standard A* function is used:

$$f(n) = g(n) + h(n)$$

1.3.2 Scaled f (for Strategies 1, 2 and 3 with scale_f = True)

If a scaling factor is applied, the cost-to-come $g(n)$ is scaled by a term that depends on the current loss $h(n)$ relative to the target loss T_L. This essentially prioritizes nodes with a loss close to the target.

$$f(n) = \max\left(0, 1 - \frac{T_L}{h(n)}\right) g(n) + h(n)$$

2 Update Strategies for Step Cost Δg

The step cost $\Delta g(n_{parent} \to n)$ for moving from a parent node n_{parent} to a neighbor node n depends on the parameter update_strategy.

Let $h_{parent} = h(n_{parent})$ and $h = h(n)$. Let h_0 be the loss of the initial node.

- C: Total number of possible quantized weight configurations.

- M: Maximum number of search iterations.
- α: Mixing factor ($0 \leq \alpha \leq 1$).
- g_{step}: A constant step value.
- k: Current iteration index.

2.1 Strategy 0: Constant Step Size

The step cost is a fixed parameter:

$$\Delta g = g_{\text{step}}$$

2.2 Strategy 1: Loss-Based Scaling (Log-Scale Configuration)

The step cost is scaled based on the relative change in loss from the parent node and the initial node.

$$\Delta g = \frac{1}{\ln(C)} \left(\alpha \frac{h}{h_{parent}} + (1 - \alpha) \frac{h}{h_0} \right)$$

2.3 Strategy 2: Normalized Constant Step Size

The step cost is a small, constant fraction of the maximum allowed iterations:

$$\Delta g = \frac{1}{M}$$

2.4 Strategy 3: Loss-Based Scaling with Iteration-Dependent Exponent

This strategy uses a complex scaling factor that is further modulated by the iteration number k using a base-10 logarithm.

$$\Delta g = \frac{1}{M \cdot \log_{10}(C)} \left(\alpha \frac{h}{h_{parent}} + (1 - \alpha) \frac{h}{h_0} \right)^{\max(1, \log_{10}(k))}$$

3 State Space

The search space is discrete, defined by the quantization of the model parameters.

3.1 Quantization Step

The quantization step δ is determined by the quantization factor $Q = \text{quantization\_factor}$:

$$\delta = \frac{1}{Q}$$

A weight w is quantized to w_q by rounding to the nearest multiple of δ:

$$w_q = \frac{\text{round}(w \cdot Q)}{Q}$$

3.2 Total Possible Configurations

The total number of possible quantized configurations, C, is used to normalize the step cost in Strategies 1 and 3. Let P be the total number of network parameters and $r_{\max} = \text{parameter\_range}[1]$ be the maximum absolute parameter value (assuming a symmetric range). The number of possible values for a single parameter is $2 \cdot r_{\max} \cdot Q + 1$.

$$C = (2 \cdot r_{\max} \cdot Q + 1)^P$$